

Data Collection & Preprocessing

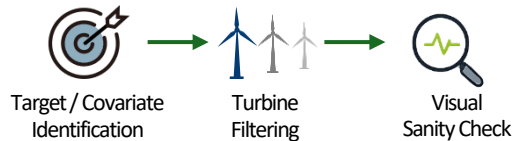
Data Sources



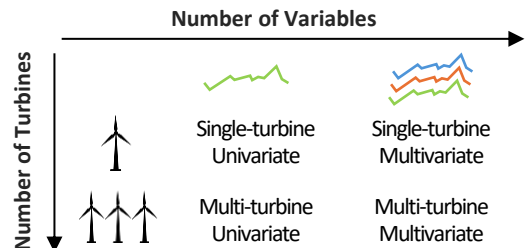
Temporal Standardization



Task-oriented Quality Control



Unified Turbine-Time-Channel Representation



Modeling

Unified Forecasting Sample

$$X \in R^{B \times N \times T \times C}$$

B:batch, N:turbines, T:look-back, C:variables

Temporal Modeling

$$(B, N, T, C) \rightarrow (B \cdot N, T, C)$$

Flatten node dimension into batch

Node-independent forecasting



Spatio-Temporal Modeling

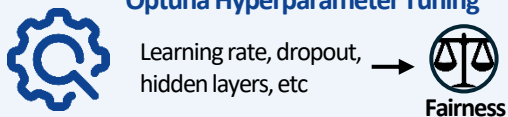
$$(B, N, T, C) \rightarrow (B, N, T, C)$$

Preserve node dimension

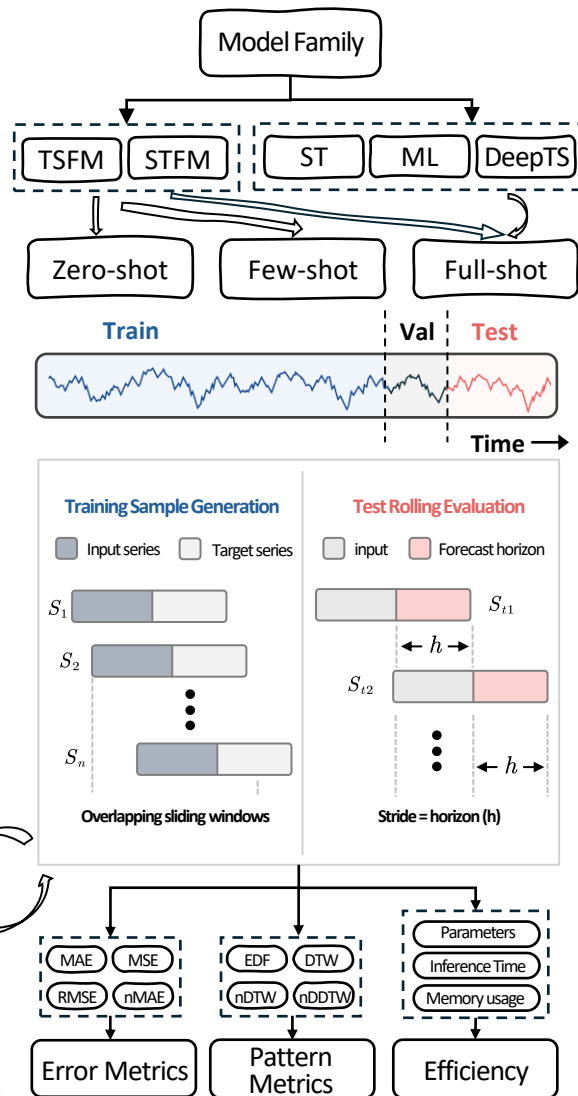
Cross-turbine Dependency modeling



Optuna Hyperparameter Tuning



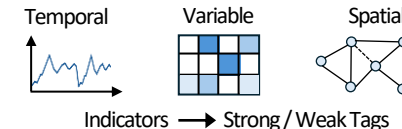
Evaluation



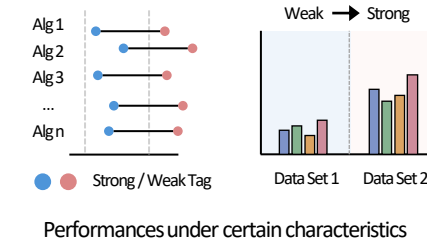
Benchmarking & Diagnostics

Analysis / Diagnostics

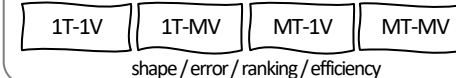
Structure Indicators



Performance Comparison



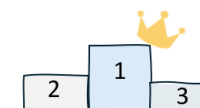
Task-group Diagnostics



Benchmarking

Error Ranking

MAE / RMSE / nMAE



Shape Ranking

DTW / nDTW / nDDTW

